TEXORA · LECTURE SHEET

The Lap Former

How loose slivers from the card and draw frame are converted into a dense, uniform lap — the only feed form the comber will accept — by the sliver lap machine, the ribbon lap machine and their modern combined lap former.

TE-2201 · SPINNING TECHNOLOGY-II · TOPIC 1 · LEVEL 2 TERM 2

1. Why the comber needs a lap

Combing is a batch action performed on a **fringe** of fibres clamped in the nippers of the comber. A sliver — round, soft and only a few grams per metre — cannot be clamped, combed and pieced up reliably: the nipper jaws would crush it unevenly, the fibres would not lie side by side in a sheet, and the feed roller could not advance it by an exact fraction of a millimetre each cycle. The feed material must therefore first be rebuilt as a **lap**: a flat, continuous sheet of fibres of fixed width (commonly about 230–300 mm), fixed weight per metre (about 25–80 g/m, most often 60–75 g/m), even from edge to edge and from metre to metre, and wound under pressure into a firm roll — also called a lap — on a steel lap rod. The machine that makes this conversion is the **lap former**, and the process is called **lap preparation**.

2. Objects of the lap former

- To combine a number of slivers (doubling) into one sheet, so that the irregularities of individual slivers average out and the lap is far more uniform than any single sliver.
- To give the fibres a light draft, which straightens and parallels them beyond what the draw frame achieved, so the comber cylinder meets an ordered fringe rather than a tangled one.
- To present the comber with a sheet of exact width and weight per metre, so the feed length set on the comber translates into a constant mass per nip — the foundation of an even combed sliver.
- To compress the sheet, by calender rollers and by winding under pressure, into a dense lap roll that unwinds smoothly on the comber's lap stand without sticking between layers (**lap licking**) and without torn edges.
- To keep the edges of the sheet clean and parallel: damaged or ragged edges comb badly and are a direct cause of waste and uneven sliver.

3. The two classical preparation methods

Traditional lap preparation uses **two machines in series**, and the lap used at the comber is the product of both.

3.1 The sliver lap machine

About 18–24 draw-frame slivers are led from cans over guide rollers through a light drafting system (draft roughly 1.5–2.5), laid side by side into a sheet, compressed by calender rollers, and wound onto a lap rod. The product — the **sliver lap** — is narrow (about 190–265 mm) and still quite bulky per unit of fibre, because the slivers have only been laid together, not truly merged. Its fibres run in distinct strands, and doubling alone has done most of the evening work.

3.2 The ribbon lap machine

Four to six sliver laps are unwound together, passed through a stronger drafting system (draft roughly 4–8, with curved drafting paths over pinned or fluted rollers that hold the fibres while they are drawn out), and wound again into the **ribbon lap**. This second drafting genuinely re-drafts the fibres into one another: strands disappear, parallelism improves, and the final lap is thinner, denser and more even. Only this ribbon lap goes to the comber.

FEATURE	SLIVER LAP MACHINE	RIBBON LAP MACHINE
Feed material	18–24 draw-frame slivers	4–6 sliver laps
Draft applied	Light, ≈ 1.5–2.5	Heavier, ≈ 4–8
Main action	Doubling and laying side by side	Re-drafting laps into one sheet
Product	Sliver lap (intermediate)	Ribbon lap (goes to the comber)
Fibre state in product	Strands still visible	Uniform, well paralleled sheet

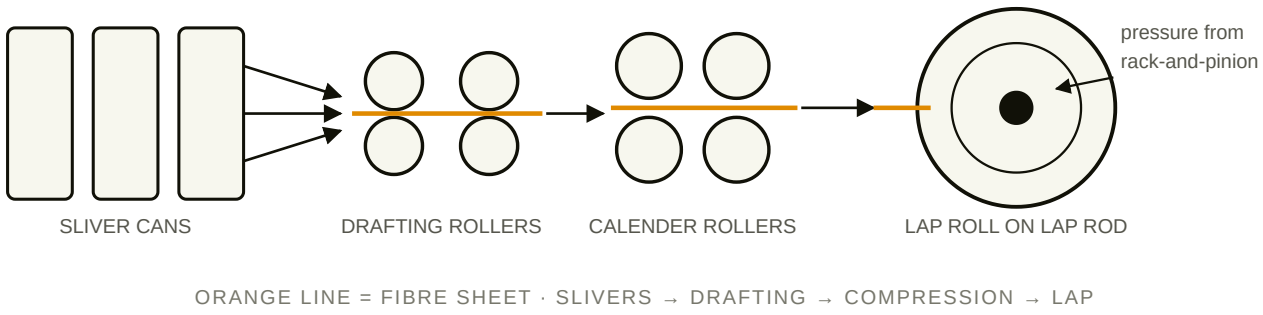


Figure 1 — Lap formation in principle (the same path is followed inside the modern combined lap former): slivers are doubled and drafted, compressed by calender rollers, then wound under pressure onto the lap rod.

4. Working principle of the modern lap former

The modern machine (for example the combined lap formers fitted to high-production combers) merges the two classical steps into one pass. Slivers from up to 24–32 cans pass over a feed table and through an autolevelling-style drafting arrangement, are condensed into a sheet of the target width, compressed between heavy calender rollers, and wound directly into the finished lap. The sequence in every machine, old or new, is identical in logic:

1. **Creeling and guiding.** Each sliver is drawn from its can without stretch, over guides that keep the slivers separate and in order; a stop motion halts the machine if a sliver breaks or a can runs empty, because one missing sliver is a permanent thin place in the lap.
2. **Drafting.** Roller pairs running at increasing surface speeds draw the bundle of slivers out — lightly in the sliver lap route, strongly in the ribbon lap, and in a controlled single pass in the modern machine. Draft straightens hooked fibres and blends the layers.
3. **Condensing and calendering.** The drafted web is guided to the exact width and squeezed between pairs of polished calender rollers, expelling air and knitting the fibres into a coherent sheet that will not fall apart in handling.
4. **Winding.** The sheet is wound onto the lap rod. Two fluted winding rollers support and drive the lap while a rack-and-pinion (or hydraulic) press holds it down with a pressure that grows as the lap builds — a hard, even package that resists lap licking on the comber yet unwinds freely.
5. **Doffing.** At the set length or weight, the machine stops, the full lap is ejected and a fresh rod inserted; the lap ends are kept clean because the comber operator must piece them up without waste.

5. Features of a modern lap former

- **Direct single-pass formation** from sliver to finished lap, removing one machine and one handling stage — the lap is never unwound and re-wound, so it suffers no extra damage.
- **Wider, heavier laps:** lap widths up to about 300 mm and lap weights up to about 80 g/m feed the widest, fastest combers; heavier laps also mean fewer lap changes at the comber.
- **Electronic drafting and monitoring:** servo-driven drafting rollers hold the set draft exactly; sensors watch sliver breaks and lap build, and the control counts production and stops on faults.
- **Uniform winding pressure** through the whole lap diameter, giving constant lap density — the cure for lap licking and for hard/soft patches that disturb comber feeding.
- **Automatic doffing and lap transport** on production lines, carrying full laps straight to the combers.

> **WORKED EXAMPLE — LAP FORMER PRODUCTION**

A lap former delivers lap at 100 m/min with a lap weight of 70 g/m, at 85 % efficiency.
Production = delivery speed × lap weight × 60 × efficiency
= 100 m/min × 70 g/m × 60 min/h × 0.85 ÷ 1000 g/kg
= 420 × 0.85 = **357 kg/h**.
The same numbers give the lap length at any target weight: a 60 kg lap at 70 g/m contains 60 000 g ÷ 70 g/m ≈ **857 m** of lap — which is why heavy laps cut comber stoppages so sharply.

6. What makes a good lap — and the faults of a bad one

FAULT	CAUSE	EFFECT AT THE COMBER
Uneven weight per metre	Draft variation, mixed sliver counts, missing doubling	Combed sliver count varies; no setting on the comber can repair it
Lap licking (layers sticking)	Too little winding pressure, hairy or over-moist lap, too heavy a lap for the fibre	Lap splits and tears on unwinding; fringe damaged, waste rises
Ragged, damaged edges	Misaligned guides, careless doffing and transport	Edge fibres escape combing; sliver edges uneven
Thin places	A sliver broke and the stop motion was ignored or by-passed	Periodic thin fringe → uneven web and sliver

Glossary

Lap — a flat sheet of fibres of fixed width and weight per metre, wound into a roll on a lap rod. **Sliver lap** — the intermediate lap made by doubling slivers on the sliver lap machine. **Ribbon lap** — the finished lap made by re-drafting several sliver laps on the ribbon lap machine. **Calender rollers** — heavy polished rollers that compress the web into a coherent sheet. **Lap rod** — the steel tube on which the lap is wound and by which it is carried to the comber. **Lap licking** — adhesion between successive layers of the lap so it tears instead of unwinding cleanly. **Doubling** — feeding several strands together so their irregularities average out.

Key points

• The comber can only work on a clamped fringe, so its feed must be a firm, even sheet — the lap.
• The lap former's objects: doubling for evenness, light draft for parallelism, exact width and weight, and firm winding without licking.
• Classically: sliver lap machine (≈ 20 slivers, draft ≈ 2) then ribbon lap machine (4–6 laps, draft ≈ 4–8).

• Modern combined lap formers do the whole conversion in one pass, with electronic draft control and uniform winding pressure.
• Production = delivery speed (m/min) × lap weight (g/m) × 60 × efficiency ÷ 1000, in kg/h.
• A bad lap cannot be corrected at the comber — lap quality sets the ceiling for combed sliver quality.

Practice

1. A lap former runs at 90 m/min delivering a 64 g/m lap at 80 % efficiency. Calculate its production in kg/h.
2. Why can a draw-frame sliver not be fed directly to the comber? Give three distinct reasons.
3. Explain why the ribbon lap step was historically necessary even though the sliver lap machine already doubles about twenty slivers.
4. A mill suffers lap licking at the comber. List three lap-former-side causes and one remedy for each.
5. A comber consumes laps weighing 55 kg each. If the lap weight is 68 g/m, how long is each lap, and why does raising lap weight reduce comber stoppages?

Answer key

1. 90 × 64 × 60 × 0.80 ÷ 1000 = 345.6 × 0.80 ≈ **276.5 kg/h**.
2. (i) The nippers cannot clamp a round, soft sliver evenly — the fringe would be irregular. (ii) The feed roller must advance an exact mass per nip; a sliver's mass per metre is far too low and too variable for a comber fringe. (iii) The fibres must already lie parallel in a sheet of fixed width so the cylinder and top comb meet an ordered fringe; a sliver is a twisted, three-dimensional strand.
3. Doubling averages irregularities but does not merge the slivers into one structure — the strands remain distinct and the parallelism is only what the draw frame gave. The ribbon lap's heavier draft (4–8) draws the fibres of several laps through one another, producing a thinner, denser, genuinely uniform sheet.
4. Causes and remedies: winding pressure too low or falling as the lap builds → restore pressure and check the press mechanism; lap too heavy/hairy for the fibre → reduce lap weight or improve fibre preparation; excessive moisture or static in the room → correct the department's humidity. (Any three consistent pairs.)
5. Length = 55 000 g ÷ 68 g/m ≈ **809 m**. A heavier lap packs more metres — and therefore more comber running time — into each lap roll, so lap changes (and their piecing-up waste and stoppage time) happen less often.